



Basic Networking For Cybersecurity

Task

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Introduction

Every organization and individual depends on networks to communicate, store, and share information. From online banking and remote work to cloud-based collaboration, nearly every digital interaction relies on secure and efficient networking. For cybersecurity professionals, understanding how networks function is essential to both identifying vulnerabilities and designing defenses that prevent compromise.

A network is not simply a collection of connected devices; it is a system of communication rules, addressing schemes, and control mechanisms. Attackers target these elements to gain access or disrupt operations, while defenders use the same knowledge to protect data and maintain resilience.

This lesson explores the foundational building blocks of networking, including IP addressing, DNS, ports, and protocols. It also introduces the concepts of network topology, architecture, and layered defense. By the end, you will understand how network design directly influences both the attack surface and the strength of your security posture.

How Devices Communicate on a Network

When you open a website, a complex chain of events happens in seconds:

1. Your device sends a request to a **DNS server** to find the IP address of the website.
2. The DNS server responds with that address.
3. Your device connects to the web server through a **port** using a **protocol** (e.g., HTTPS).
4. Data travels across multiple **layers** (physical wires, routers, and software) before reaching its destination.

Each step represents both a function and a potential security risk.

Analogy:

Think of a network like a postal system.

- The **IP address** is the destination address on the envelope.
- The **DNS** is the address book that tells you where to send it.
- The **port** is the specific office or department that receives the letter.
 - The **protocol** is the language and format used to write the letter.

Network Fundamentals

A computer network is a collection of devices, or nodes, connected to share information and resources. These connections can be wired, wireless, or hybrid, and may include servers, routers, switches, laptops, and Internet of Things (IoT) devices.

Networks are categorized by scope:

- **Local Area Network (LAN):** Connects systems within a limited area, such as an office or campus.
- **Wide Area Network (WAN):** Connects multiple LANs across cities or countries, often through the internet.
- **Wireless Networks (Wi-Fi):** Use radio signals to allow mobile access, requiring additional security controls.

For cybersecurity, understanding how networks are built and managed is critical. Every connected device, application, or user adds potential vulnerabilities. Effective defense begins with visibility knowing what systems exist, how they communicate, and which services are exposed.

The Role of IP Addresses

The Internet Protocol (IP) is the addressing system that allows devices to locate and communicate with one another across networks. An IP address identifies each device uniquely, ensuring that data packets reach the correct destination.

There are two main versions of IP addresses:

- **IPv4:** The most widely used system, consisting of four numbers separated by dots (for example, 192.168.1.1). IPv4 supports about 4 billion addresses, but global demand has nearly exhausted this supply.
- **IPv6:** A newer system that uses longer alphanumeric sequences (for example, 2001:db8::1). It provides an enormous address range and includes improved efficiency and security features.

To efficiently manage the limited IPv4 space and enhance security through isolation, networks use subnetting and Classless Inter-Domain Routing (CIDR). Subnetting divides a large IP range into smaller subnetworks (subnets) using a subnet mask, which specifies how many bits are used for the network portion versus hosts. For example, the range 192.168.1.0/24 (subnet mask 255.255.255.0) allows 254 usable hosts on one subnet, while /25 splits it into two subnets of 126 hosts each.

CIDR notation (e.g., /24) enables flexible address allocation beyond rigid IPv4 classes (A/B/C), conserving addresses and supporting hierarchical routing. In cybersecurity, subnetting limits the blast radius of attacks: If a device in one subnet is compromised, it cannot easily reach others without routing approval.

Attackers exploit poor subnetting by flooding broadcasts across large ranges (broadcast storms) or using tools to enumerate subnets via ICMP pings. Defenders calculate subnets for segmentation, deploy Virtual LANs (VLANs) to enforce them logically on switches, and use access control lists (ACLs) to block inter-subnet traffic unless authenticated.

Attackers often exploit IP-based vulnerabilities through techniques like IP spoofing, where they disguise their address to appear legitimate, or DDoS (Distributed Denial-of-Service) attacks, which overwhelm networks with traffic from multiple sources.

To counter these risks, defenders:

- Segment internal networks to restrict movement between systems.
- Use firewalls to filter and inspect traffic based on IP ranges.
- Implement access control lists (ACLs) to allow only trusted sources.
- Monitor for unusual traffic patterns that indicate scanning or brute-force activity.

An organization that understands its IP landscape can identify which systems are exposed to the public and which must remain isolated.



Take note

Many successful attacks begin with network reconnaissance; mapping IP addresses to see what services are running. This is why external IP visibility should always be limited, and internal IPs should be protected through proper segmentation.

Understanding DNS

The Domain Name System (DNS) functions as the internet's address book, translating easy-to-read domain names into numerical IP addresses. When a user types a web address into a browser, DNS servers look up the corresponding IP so that the request can be routed to the correct destination.

Without DNS, users would need to remember strings of numbers for every website. However, because DNS is so fundamental, it is also a powerful target for attackers. Common DNS attacks include:

- **DNS Spoofing:** Injecting false records to redirect users to malicious websites.
- **DNS Tunneling:** Using DNS queries to exfiltrate data or command remote systems.
- **Domain Hijacking:** Taking over a registered domain to impersonate an organization.

For internal networks, private DNS servers resolve local hostnames to private IPs (e.g., 10.0.0.0/8 ranges), often paired with DHCP (Dynamic Host Configuration Protocol) to automatically assign IPs, masks, and gateways to devices. Misconfigured DHCP can lead to rogue servers handing out fake IPs, enabling MitM attacks.

Defensive measures include implementing DNSSEC (DNS Security Extensions) to validate DNS responses, encrypting DNS queries with DNS over HTTPS (DoH) or DNS over TLS (DoT), and maintaining strict access

control to DNS servers. Monitoring DNS logs can also reveal early indicators of compromise, such as unexpected outbound requests to suspicious domains.



Extra resource

Learn more about how DNSSEC adds authentication and integrity to domain resolution: [DNSSEC Overview – ICANN](#)

Ports and Protocols

Every device on a network communicates through ports, which act as numbered channels that data uses to enter or exit a system. Each port is associated with a protocol, which defines how data is formatted, transmitted, and verified.

There are 65,535 ports in total, divided into:

- **Well-known ports (0–1023):** Used for standard services like HTTP (80), HTTPS (443), FTP (21), and SSH (22).
- **Registered ports (1024–49151):** Assigned to specific applications such as SQL databases or messaging services.
- **Dynamic or private ports (49152–65535):** Used temporarily for client connections.

Protocols serve as the language of communication between devices. Common examples include:

- **TCP (Transmission Control Protocol):** Ensures reliable, ordered delivery of data. Used by web and email services.
- **UDP (User Datagram Protocol):** Faster but without reliability checks. Common in streaming and gaming.
- **HTTP/HTTPS:** Handle web traffic, with HTTPS using encryption for security.
- **SMTP, IMAP, POP3:** Govern email transmission and retrieval.
- **FTP/SFTP:** Enable file transfers, with SFTP providing encryption.

Beyond core protocols, Network Address Translation (NAT) and Port Address Translation (PAT) are crucial for conserving public IPs and adding a layer of obscurity. NAT rewrites IP headers to map multiple private internal addresses (e.g., 192.168.1.10) to a single public one (e.g., 203.0.113.1), hiding the internal topology from external scanners. PAT extends this by multiplexing ports, allowing thousands of internal devices to share one public IP (e.g., internal port 12345 maps to external 80).

In cybersecurity, NAT acts as a basic firewall but can be bypassed via UPnP (Universal Plug and Play) misconfigurations that auto-open ports. Attackers use port forwarding exploits to tunnel malicious traffic inward. Defenses include disabling UPnP, using stateful NAT on routers to track connections, and logging translations for anomaly detection.

For wireless networks, protocols like WPA2 and WPA3 secure Wi-Fi links. WPA2 uses PSK (Pre-Shared Key) with TKIP/AES encryption, but is vulnerable to KRACK (Key

Reinstallation Attacks) that replay handshakes. WPA3 introduces SAE (Simultaneous Authentication of Equals) for stronger key exchange and forward secrecy, resisting offline dictionary attacks. Wireless-specific threats include deauthentication floods (DoS via fake disconnects) and evil twin APs (rogue hotspots mimicking legitimate ones). Defenders enable WPA3, hide SSIDs, use certificate-based EAP for enterprise auth, and deploy wireless intrusion detection to scan for unauthorized signals.

Utility protocols like ICMP (Internet Control Message Protocol) and SNMP (Simple Network Management Protocol) are often overlooked but heavily exploited. ICMP enables diagnostics (e.g., ping for reachability) but fuels DoS via echo floods. SNMP monitors devices but versions 1/2 use plaintext community strings (e.g., "public") for read/write access, leaking configs. Defenses: Rate-limit ICMP, disable echoes externally, and mandate SNMPv3 with SNMPv3 authentication/encryption.

Cyber attackers use **port scanning** to discover open or vulnerable ports. An unmonitored port can act as a backdoor into a system. For example, leaving Remote Desktop Protocol (RDP) open to the internet has led to countless ransomware infections.

Network administrators close unnecessary ports, configure firewalls to control incoming and outgoing traffic, and monitor ports for suspicious behavior.



Take note

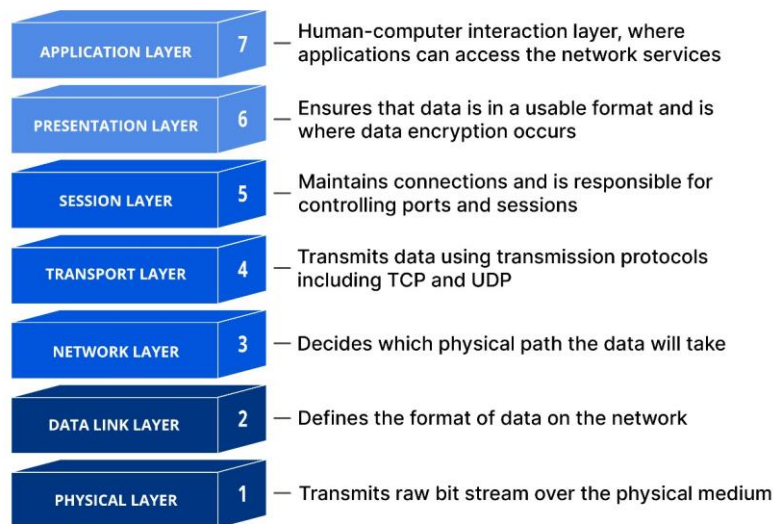
Ports and protocols always work together. A port without a protocol is simply an open doorway. Understanding which ports are active and what protocols they serve is fundamental to securing a network.



Code hack

Tools such as [Nmap](#) or [Zenmap](#) allow security professionals to scan for open ports and check for vulnerabilities. Ethical scanning in controlled environments helps identify weaknesses before attackers exploit them.

Network Layers and the OSI Model



Cloudflare. (n.d.). What is the OSI Model? Cloudflare. Retrieved November 10, 2025, from <https://www.cloudflare.com/learning/ddos/glossary/open-systems-interconnection-model-osi/>

The Open Systems Interconnection (OSI) model is the standard framework used to understand how data travels from one device to another. It consists of seven layers, each responsible for a specific part of the communication process:

1. **Physical Layer:** Manages physical hardware, cables, and signals. Attacks may involve tampering with routers, installing rogue devices, or disrupting connections.
2. **Data Link Layer:** Handles communication between devices on the same network using MAC addresses. Common attacks include MAC spoofing and ARP poisoning.
3. **Network Layer:** Determines how packets travel across networks using IP addressing. IP spoofing and routing manipulation are frequent threats.
4. **Transport Layer:** Controls data delivery using TCP and UDP. Defenders use rate limiting to prevent denial-of-service (DoS) attacks.
5. **Session Layer:** Manages connections between devices. Secure sessions prevent hijacking and unauthorized re-use of credentials.
6. **Presentation Layer:** Formats and encrypts data before transmission. Weak encryption or unvalidated input can compromise security.

7. **Application Layer:** Supports end-user services such as email, web browsing, and file transfer. This layer faces the most attacks, including SQL injection and cross-site scripting.

Each layer builds upon the one below it, and a compromise at any layer can disrupt the entire communication chain. Security professionals apply defense-in-depth principles to ensure protection across every layer: firewalls and IDS systems at the network layer, encryption at the presentation layer, and secure coding practices at the application layer. For example, If users cannot reach a website, an issue might exist at Layer 3 (routing) or Layer 7 (application). Understanding OSI helps defenders isolate the problem and apply the right fix efficiently.



Extra resource

For a clear visual explanation, explore: [OSI Model Explained – Cloudflare Learning Center](#)

Network Topology and Architecture

Network topology describes how devices are connected, while network architecture refers to how they are organized and protected. Both influence how efficiently data moves and how well the network resists attack.

Common topologies include:

- **Star topology:** Devices connect to a central switch or hub. Simple to manage but vulnerable if the central device fails.
- **Bus topology:** Devices share one communication line. Inexpensive but prone to congestion and single-point failure.
- **Ring topology:** Data travels in a loop between devices. Each device acts as a repeater, but a single break can disrupt the ring.
- **Mesh topology:** Every device connects to multiple others, offering redundancy and resilience.
- **Hybrid topology:** Combines several designs for performance and flexibility.

Beyond structure, architecture determines **how access is controlled** and **how data is segmented**. Key architectural security elements include:

- **Segmentation:** Dividing a network into subnets or zones to contain breaches.
- **Firewalls and Demilitarized Zones (DMZs):** Placing public-facing services, like web servers, in a controlled zone separate from internal systems.

- **Zero Trust:** Verifying every request, even from inside the network, based on identity and context.
- **Virtual Private Networks (VPNs):** Securing remote connections with encryption.

To implement segmentation without rewiring, VLANs create logical subnetworks on a single physical switch, tagging Ethernet frames with VLAN IDs (IEEE 802.1Q). For instance, VLAN 10 for finance and VLAN 20 for guests prevent cross-talk, even on shared trunks (links carrying multiple VLANs). VLAN hopping attacks exploit double-tagging or switch spoofing to jump boundaries.

Defenses: Configure native VLANs securely, disable dynamic trunking, and use private VLANs to isolate ports within a VLAN.

Architectural intermediaries like proxy servers enhance control. Forward proxies sit between clients and the internet, caching content and filtering requests (e.g., blocking malware domains). Reverse proxies front servers, load-balancing traffic and shielding origins from direct exposure. Both inspect for threats but can be abused for data leaks if unauthenticated. Defenses: Enforce HTTPS inspection, authenticate users via Kerberos/NTLM, and rotate proxy IPs to evade blacklists

A business may keep its email and payroll servers on separate segments. If attackers breach the email system, segmentation prevents them from accessing payroll data. For beginners, visualizing network design as layers of physical and logical protection helps clarify why structure matters as much as technology.



Take note

Good network design does not depend on expensive hardware. It depends on understanding how to separate, control, and monitor the flow of information. Even small networks can be made secure with proper architecture.

Network Hardening and Secure Configuration

A secure network begins with devices that are configured according to recognized security baselines. Hardening ensures that routers, switches, firewalls, access points, and servers operate only with the features necessary for business operations, reducing the number of pathways an attacker can exploit. Because network devices often ship with default settings optimized for convenience, attackers frequently target them first.

Hardening typically involves disabling unused ports and services, enforcing encrypted management protocols, applying strong authentication, and removing default credentials. For example, many routers still ship with administration enabled over HTTP using basic credentials like “admin/admin.” Attackers routinely scan the internet for these interfaces. When an organization enables HTTPS, restricts access to a secure management VLAN, and enforces strong credentials, these attacks are no longer effective.

Replacing insecure protocols with secure alternatives is also essential. For example, let’s say we have a small business using Telnet for switch administration. Telnet sends credentials in plain text and exposes them to interception. When the business replaces Telnet with SSH, all administrative communication becomes encrypted, preventing this type of eavesdropping.

Hardening also applies to simple settings that are often overlooked. For example, a switch using the default SNMP community string “public” can reveal device information to anyone who discovers it. Changing this to SNMPv3 with authentication and encryption prevents unauthorized access.

How Professionals Do This in Practice

Cybersecurity professionals apply hardening through a structured workflow:

1. **Reviewing current configurations:**
They export device configurations, review active services, and identify insecure defaults.
2. **Using security baseline templates:**
Templates specify required settings, such as “SSH only,” “disable Telnet,” and “restrict management access.”
3. **Applying changes in controlled windows:**
All changes are documented, reviewed by a Change Advisory Board, and applied during maintenance windows.
4. **Testing and validation:**
They ensure connectivity remains stable, scan exposed ports, and validate that all security controls function correctly.
5. **Continuous monitoring:**
Tools monitor configuration drift and flag unauthorized or insecure changes.

Consistent hardening reduces exposure, enforces predictable behavior, and establishes a strong foundation for all other network security controls.

Secure Routing and Path Control

Routing determines how data moves through a network. If routing is unprotected or misconfigured, attackers can influence traffic paths, redirect data, or disrupt communication. Secure routing ensures that only trusted devices can exchange routing information and that routing decisions are based on authenticated, authorized updates.

Attackers sometimes attempt route poisoning or route injection, where they introduce false routing information to divert traffic. For example, let’s say we have an internal OSPF network with authentication disabled. An attacker on the same segment could impersonate a router and announce a more attractive route. The result is that traffic may be rerouted through a system the attacker controls. Enabling OSPF authentication prevents this because only authorized routers can exchange routing updates.

Route filtering also plays a critical role. For example, let’s say a remote office advertises local subnets to headquarters. Without filtering, it might accidentally advertise networks it should not expose. Using prefix lists and route maps, the organization ensures only approved networks are exchanged.

Secure routing extends into cloud environments. A misconfigured cloud routing table can accidentally send internal traffic through a public gateway, exposing workloads. Regular inspection of cloud route tables and VPC peering rules prevents such exposures.

How Professionals Do This in Practice

Cybersecurity and network professionals secure routing through structured activities:

1. **Identifying active routing protocols:**
They map where OSPF, EIGRP, BGP, or static routes are used.
2. **Enabling routing authentication:**
Protocol authentication (e.g., OSPF MD5) ensures only trusted routers exchange updates.
3. **Applying route filtering:**
Prefix lists control which networks can be advertised or accepted.
4. **Reviewing routing tables regularly:**
Professionals confirm no unexpected networks appear and validate traffic paths.
5. **Monitoring for routing changes:**
Alerts flag unexpected route shifts or neighbor drops.
6. **Testing routing integrity:**
They use traceroute, packet captures, and failover tests to ensure routes behave as expected.

Securing routing ensures that traffic follows predictable, trustworthy paths and prevents attackers from manipulating the flow of information.

Monitoring and Defense

Once a network is designed, it must be continuously monitored. Attackers constantly probe for weaknesses, and early detection can mean the difference between a minor incident and a major breach.

Network monitoring tools such as Wireshark and tcpdump allow professionals to inspect traffic at the packet level. Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) analyze network behavior to identify suspicious activity. Security Information and Event Management (SIEM) platforms collect logs from multiple systems, providing centralized analysis and alerts.

Other important practices include:

- **Network Access Control (NAC):** Ensuring only approved devices can connect.
- **Regular audits:** Reviewing configurations and permissions for changes.
- **Patch management:** Keeping routers, firewalls, and firmware up to date.
- **Encryption:** Securing data in transit with TLS or IPsec.

Enhance visibility with protocol-specific monitoring: Filter Wireshark captures for ARP requests to detect poisoning, or SNMP traps for unauthorized device polls. For wireless, tools like Kismet scan spectrum for rogue APs. Integrate NAC with 802.1X for port-based auth, dynamically assigning VLANs based on device posture (e.g., patched OS). Regular audits include subnet scans (e.g., via Angry IP Scanner) to map exposed ranges and proxy logs for policy violations.



A secure network is one that is visible, monitored, and actively managed.

Practical task

This task requires you to use specific online tools to perform basic network reconnaissance on a publicly visible target, [microsoft.com](https://www.microsoft.com), and then apply your knowledge of CIDR notation to analyze the resulting IP address.

Instructions

Follow these three sequential steps, using the specified online tools, and record your findings directly into the Submission Table provided below.

Step 1: DNS Lookup (Finding the Target)

1. Go to the online tool DnsChecker.org.
2. In the search bar, type the target domain: microsoft.com.
3. Ensure the record type is set to A (IPv4 Address).
4. Record one of the returned IPv4 addresses in the Submission Table.

Step 2: CIDR Analysis (Determining the Range)

1. Go to the online tool Subnet-Calculator.com.
2. In the IP Address field, enter the IPv4 address you found in Step 1.
3. Ensure the Subnet Mask / CIDR field is set to /24 (the default for a common Class C network).
4. Record the Network Address and the Broadcast Address for the /24 subnet in the Submission Table.

Step 3: Basic Port Scanning (Checking Exposure)

1. Go to the online tool YouGetSignal.com's Port Forwarding Tester.
2. In the IP Address or Hostname field, enter the IPv4 address you found in Step 1.
3. Check the status of the following three standard ports one by one: Port 80 (HTTP), Port 443 (HTTPS), and Port 22 (SSH).
4. Record the resulting status (Open, Closed, or Filtered) for each port in the Submission Table.

Submission Requirement

Copy the Submission Table below into a word processing document (such as Google Docs or Microsoft Word). Fill in all the required information based on your tool usage. Save and submit your completed document as a PDF file.

Submission Table

Metric	Tool Used	Result
Target Domain	N/A	microsoft.com
Public IPv4 Address Found	DnsChecker.org	150.171.109.119
/24 Network Address	Subnet-Calculator.co m	150.171.109.0
/24Broadcast Address	Subnet-Calculator.co m	150.171.109.255
Status of Port 80 (HTTP)	YouGetSignal.com	Open
Status of Port 443 (HTTPS)	YouGetSignal.com	Open
Status of Port 22 (SSH)	YouGetSignal.com	Filtered

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.



Share your thoughts

Please take some time to complete this short feedback [form](#) to help us ensure we provide you with the best possible learning experience.

Reference list

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